UNIVERSITÉ DE GENÈVE
GENEVA SCHOOL OF ECONOMICS AND MANAGEMENT

Introduction to Macroeconomics

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Solution: Problem Set 2

2026

1 Consumer Price Index

Solution 1

We first calculate the CPI in 2004 and 2005, taking 2004 as the base year. Then, we calculate the inflation rate in 2005 using the two CPIs obtained.

1. We construct a representative and fixed baskets of goods.

We take the 2004 quantities to construct the fixed basket of goods (100 heads of cauliflower, 50 bunches of broccoli and 500 carrots) : $Q_c = 100$, $Q_b = 50$, $Q_{cr} = 500$.

2. We find the price of each good in each year

On a given year, the price of a good can be calculated by dividing the amount spent on that good by the quantity of good bought.

$$P_{c,2004} = \frac{200}{100} = 2 \qquad P_{b,2004} = \frac{75}{50} = 1.50 \qquad P_{cr,2004} = \frac{50}{500} = 0.10$$

$$P_{c,2005} = \frac{225}{75} = 3 \qquad P_{b,2005} = \frac{120}{80} = 1.50 \qquad P_{cr,2005} = \frac{100}{500} = 0.20$$

3. We calculate the cost of the representative fixed consumption basket on each year

2004 : $P_{c,2004} \cdot Q_c + P_{b,2004} \cdot Q_b + P_{cr,2004} \cdot Q_{cr} = 2 \cdot 100 + 1.50 \cdot 50 + 0.10 \cdot 500 = 325$

2005 : $P_{c,2005} \cdot Q_c + P_{b,2005} \cdot Q_b + P_{cr,2005} \cdot Q_{cr} = 3 \cdot 100 + 1.50 \cdot 50 + 0.20 \cdot 500 = 475$

4. We choose a base year and set the cost of the fixed consumption basket to 100 in that year (in our case, 2004). We set the cost of the fixed consumption basket to 100 on that year and we calculate the CPI for year *X* as follows :

$$CPI_X = \frac{\textit{cost of fixed basket on year X}}{\textit{cost of fixed basket on base year}} \cdot 100$$

Thus :

$CPI_{2004} = \frac{325}{325} \cdot 100 = 100$ (the CPI for the base year is always 100, obviously...)

$CPI_{2005} = \frac{475}{325} \cdot 100 = 146$

5. The inflation rate is given by the percentage change in the CPI

$$\textit{Inflation rate}_{2005} = \frac{CPI_{2005} - CPI_{2004}}{CPI_{2004}} \cdot 100\% = \frac{146 - 100}{100} \cdot 100\% = 46\%$$

Solution 2

Answer (c) The nominal wage of the employee increased by $\frac{200}{2500}$ = 1.08. Prices increased by the same amount and thus her real wage did not change.

Solution 3

⇒ **Answer (c)**

To evaluate the cost of the representative basket on year X we multiply each quantity by its price in year X, and then we sum them all :

2000 : $10 \cdot 2 + 1 \cdot 50 + 200 \cdot 1 + 50 \cdot 2 + 10 \cdot 100 = 1370$

2001 : $10 \cdot 1 + 1 \cdot 60 + 200 \cdot 1 + 50 \cdot 3 + 10 \cdot 120 = 1620$

2002 : $10 \cdot 3 + 1 \cdot 70 + 200 \cdot 1 + 50 \cdot 4 + 10 \cdot 150 = 2000$

Therefore, the CSPI for these three years, taking 2000 as the base year :

$$2000 \quad : \qquad \frac{1370}{1370} \cdot 100 = 100$$

$$2001 \quad : \qquad \frac{1620}{1370} \cdot 100 = 118$$

$$2002 \quad : \qquad \frac{2000}{1370} \cdot 100 = 146$$

Therefore, we see that the CSPI indicates that the cost of living for students has increased along the years. However, in this treatment of the data, we have considered that the basket for the three years is the same, that is, we have implicitly assumed that the calculators are the same in 2000, 2001 and 2002. Their price has certainly increased, contributing to the rise in the CPI observed. However, the calculator in 2000 is not the same as the one in 2001 and the one in 2002 : the quality has increased. Part of the increase of the price is due to the improvement in the quality of the calculator. As a result, the CSPI overestimates the cost of living by not making an adjustment for the increase in product quality.

In addition, the CSPI also overestimates the cost of living due to the substitution bias, since the students could have increased their consumption of coffee as a substitute to high-caffeine soda as the price of the latter increased.

2 Labor market

Solution 4
Solution : (d)

Note that (d) implies that (a), (b) and (c) are correct. In these kind of exercises, if (a), (b) and (c) are correct, then (d) is correct, which means that we have four correct statements. However, as (d) is the statement that incorporates all the other ones, it is the one that should be chosen.

At the equilibrium, the labor (here the units are in number of workers) is $L = 100$ and the wage (the price of labor) is $w = 10$. The labor market is analyzed in the same way as any other contract :

— A labor demand L^D, that originates from the profit-maximizing behavior of firms

— A labor supply L^S, that originates from the utility-maximizing behavior of workers

— An equilibrium, in which the condition supply=demand is fulfilled.

(a) True :

The profit-maximizing behavior of firms implies that the real wage must be equal to the marginal productivity of labor. Therefore, all the points of the labor demand curve (or line, in this case) are points where the marginal productivity of labor is equal to wage. As we can see, as the amount of labor (which is a factor of production) increases, the wage and thus the marginal productivity of labor decreases. This is consistent with diminishing marginal returns to labor.

Therefore, since the equilibrium is a point that belongs to the demand curve, the condition

$$w = \textit{marginal productivity of labour}$$

is satisfied. Since the real wage is $w = 10$ at the equilibrium, the marginal productivity of labor is equal to 10.

(b) True :

If a minimum wage is set at a wage $\underline{w}$ whose value is less than 10, then the wage cannot be lower than $\underline{w}$, but it still can be 10. As the market tends "naturally" towards the equilibrium, the wage will be equal to 10.

If the minimum wage was above 10, structural unemployment would appear, due to an excess supply of labor. Let's assume, for example, that the minimum wage is set at 15. The wage then would have to be greater or equal to 15. At any point above w=15 we have an excess supply, and therefore the market would "naturally" tend to go towards the equilibrium. But it cannot reach the equilibrium, as wages below 15 are forbidden by the law. Therefore, the wage would be equal to 15, we would have an excess supply of work, and we would have structural unemployment.

(c) True :

The supply of labor is based on the utility-maximising behavior of workers. The reservation wage is the (hourly) minimum wage requested by the person to participate in the labor market. For wages below the reservation wage, the labor supply is zero. The reservation wage is determined by :

— Income spouse, "and legacies" family, the state transfers and other non-state aid, the opportunity to work in black,

— The existence or non-cash fixed costs (e.g. child care) and non-monetary (e.g. travel time)

— The system of transfers and direct taxation.

We can imagine that workers are ordered on the horizontal axis according to their reservation wage. To the left of axis, unmarried young and urban people (no income spouse or fixed costs (non) monetary) are willing to work for low wages, that is to say they have a low salary reserve. To the right of the axis, we have mothers of large families with husband and CEO living in the countryside (high income spouse, fixed costs (non) monetary). So they will work only if the salary is very high, that is to say, they have a high reservation wage. Therefore, they will only work if the wage is very high.

However, the wage on the market is set by the interaction of supply and demand, and it is unique for this market. Therefore, the workers who are on the left of the equilibrium point get a higher wage than the one they would be ready to work for (the value of the wage is higher than their marginal productivity of labor). Only the worker who is exactly at the equilibrium point gets a wage that is equal to his/her reservation wage. In this case, since at equilibrium L = 100, only the 100th worker earns the value of his/her own reservation wage.

Solution 5

Solution : (1)

The labor force is made up of the number of employed workers plus the number of unemployed workers. The unemployment rate is calculated by dividing the number of unemployed workers by the total labor force. Please note that we are only interested in the labor force : retirees and those not seeking employment (e.g. severely disabled.) are not counted in the labor force.

Therefore, if the number of employed workers falls, the unemployment rate tends to rise. In all cases, if the number of unemployed workers does not change but the number of employed workers falls, the labor force becomes smaller and therefore the unemployment rate rises.

However, in this problem we see that even if the number of employed workers falls, the unemployment rate also falls. Is there any explanation for this ? Yes : if the number of employed workers decreases and the number of unemployed workers also falls, the decrease in the labor force is not only due to the decrease in the number of employees but also due to the decrease in the number of unemployed workers. In any case, if the fall in the number of unemployed workers is large enough with respect to the fall in the number of employed workers, it is possible to observe simultaneously a fall in the number of employed workers and the unemployment rate.

Let's illustrate this for this example :

Let's define X as the number of unemployed workers in 2009 and Z as the number of unemployed workers in 2010.

$$\begin{aligned}
\textit{unemployment rate} &= \frac{\textit{number of unemployed workers}}{\textit{labor force}} \cdot 100\% = \\
&= \frac{\textit{number of unemployed workers}}{\textit{number of unemployed workers} + \textit{number of employed workers}} \cdot 100\%
\end{aligned}$$

Therefore, for 2009 :

$$
\begin{aligned}
\textit{unemployment rate} &= \frac{X}{X + 137,738,000} \cdot 100\% = 6.4\% \\
\frac{100}{6.4}X &= X + 137,738,000 \\
\frac{100 - 6.4}{6.4}X &= 137,738,000 \\
14.625X &= 137,738,000 \\
X &= \frac{137,738,000}{14.625} = 9,417,983
\end{aligned}
$$

And for 2010 :

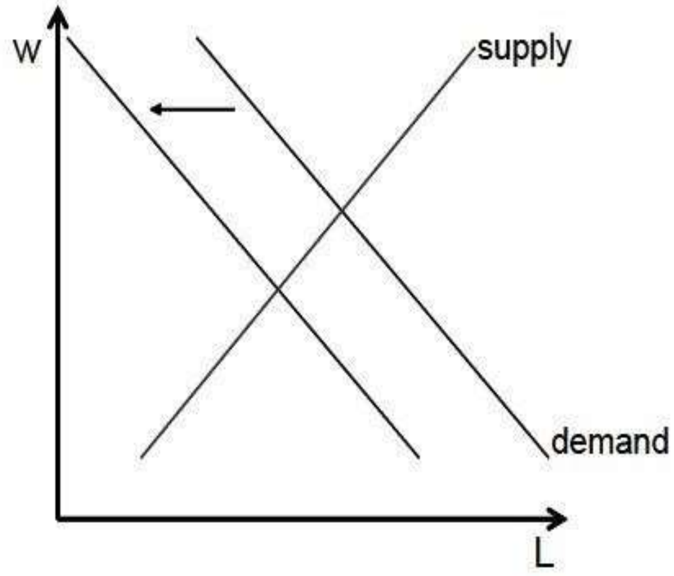
$$
\begin{aligned}
\textit{unemployment rate} &= \frac{Z}{Z + 137,478,000} \cdot 100\% = 6.2\% \\
\frac{100}{6.2}Z &= Z + 137,478,000 \\
\frac{100 - 6.2}{6.2}Z &= 137,478,000 \\
Z &= 137,478,000 \\
Z &= \frac{137,478,000}{14.625} = 9,087,032
\end{aligned}
$$

In this case, the number of employed workers fell by 260,000 (from 137,738,000 to 137,478,000), while the number of unemployed workers fell by 330,951 (9,417,983 to 9,087,032).This caused the labor force to shrink by 590,951. The decrease in the number of unemployed workers is large enough to cause the drop in the unemployment rate, even if the labor force has declined too.

Solution 6

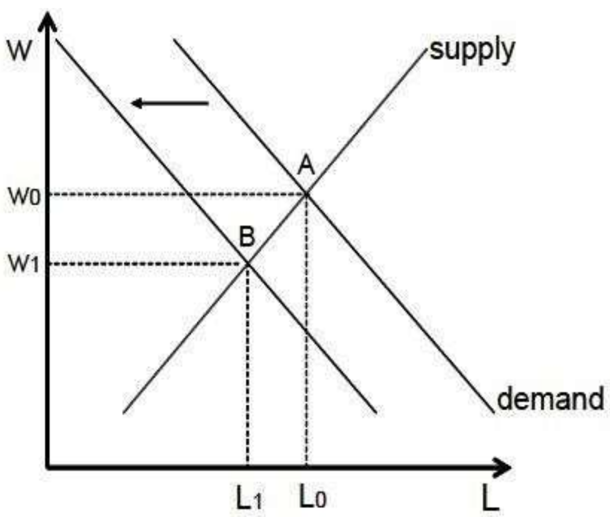
Solution 6.1 :

Workers become less productive, and therefore firms wish to hire less workers. The labor demand curve shifts to the left (for any given wage, the firms hire less workers, and thus L is smaller).



Solution 6.2 :

What happens is that we move from equilibrium A to equilibrium B. The employment level falls down from L_0 to L_1, and so does the real wage (from w_0 to w_1). This is a response to a decrease in productivity.



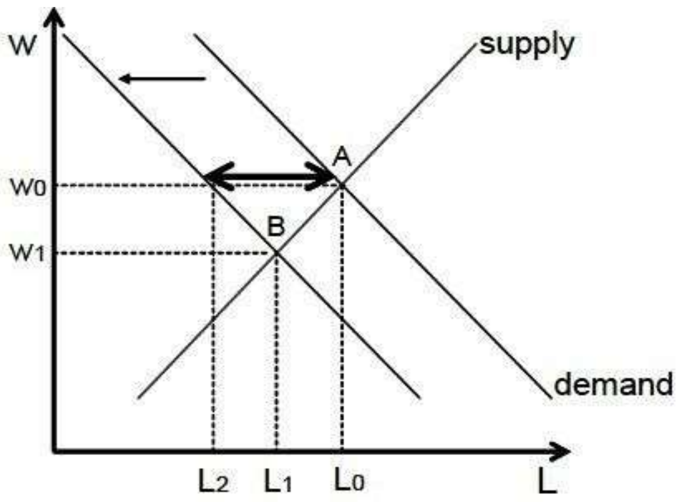
Since *A* and *B* are equilibrium points, the demand is equal to the supply in both cases. Therefore, in both cases, the number of jobs that the companies offer are equal to the number of people who want a job. There is no structural unemployment.

Reminder : structural unemployment is the unemployment that occurs when the number of jobs available in the labor market is insufficient to provide a job for everyone who wants to work.

Solution 6.3 :

In this case, w cannot go down from w_0 to w_1, and therefore B is not attainable. The wage stays at w_0 and, since the demand has shifted to the left, an excess supply appears, which generates structural unemployment. This excess supply is represented by the double arrow :

the number of people who wish to work (L_0) is higher than the number of jobs offered by firms at this wage (L_2).



In general, when there is an excess supply or an excess demand, there is a difference between the amount of good the producers are willing to supply and the amount of good the consumers are willing to buy. In such a case, the quantity that is exchanged is the smaller of the two. Here, as labor is treated like a good, and we can apply the same principle : the level of employment is the smaller value between the number of people who wish to work and the number of jobs offered by firms. Therefore, the level of employment is L_2. Because at w_0 there are agents that want to work but cannot ($L_0 > L_2$), there is structural unemployment.

If there is no union agreements, the shift in the demand curve would result in a change from equilibrium A to equilibrium B. The employment level would fall to L_1 and the wage would also fall to w_1. But if the wage cannot go down, the wage stays at its initial value (w_0) but the level of employment falls even more than in absence of the union agreements, as it falls to L_2. Therefore, according to this model, those who are lucky enough to work are not worse off, as the wage is the same as before the shift in demand. However, unions agreements lead to the emergence of structural unemployment to a lower level of employment than in absence of these union agreements.

3 Additional exercises (will not be discussed in the seminar)

Solution 7

⇒**Answer (b)**

Seeing as the level of prices has changed over the two years (the CPI is different across the two years), we cannot use the increase in nominal income as a measure of a change in purchasing power (remember that purchasing power depends on both the nominal income and the price level). One way to evaluate the change in purchasing power is to calculate the real income in both periods (income at the price level of a reference year), and to compare the obtained values.

We can calculate the real income for any year X (relative to the reference year on which the CPI is based), by dividing the nominal income in year X by the CPI in year X and multiplying by 100 (we need to multiply by 100 since the CPI in the reference year is always 100).

$$real\ income_{this\ year} = nominal\ income_{this\ year} \cdot \frac{100}{CPI_{this\ year}} = 31000 \cdot \frac{100}{169} = 18343,20$$
$$real\ income_{last\ year} = nominal\ income_{last\ year} \cdot \frac{100}{CPI_{last\ year}} = 19000 \cdot \frac{100}{122} = 15573,77$$

Thus, seeing as the real income has increased, it must be that the purchasing power has increased as well.

Solution 8

Solution : (d)

- Seasonal unemployment is due to variations in activity in certain sectors of the economy throughout the year (for example tourism sector, wine industry, etc..)
- Structural unemployment is due to wage rigidities (ex minimum wages) and qualifications.
- Cyclical unemployment is due to the short-term ups and downs of the business cycle.
- Frictional unemployment is due to the fact that it takes time to search for a job.

Solution 9

⇒ Answer (c)

Efficiency wages, minimum wages, and labour union bargaining are all part of the wage rigidities, which are at the root structural unemployment.